COIMBATORE INSTITUTE OF TECHNOLOGY

(GOVERNMENT AIDED AUTONOMOUS INSTITUTION)

CIVIL AERODROME POST, COIMBATORE-641014.

PLACEMENT AND TRAINING CELL

PLACEMENT YEAR 2021-2022

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COMPANY TYPE	: DREAM
JOB DESIGNATION	: Member Technical (Quality and Testing Engineer)
SALARY (CTC)	: 34,25,000 - 35,25,000
INTERN OFFERED ?	: NO
BOND	: NO
HAVE YOU PLACED ?	: YES

COMMENTS ON SELECTION PROCESS :

ROUND 1 :

Platform: HackerRank

Round details: 14 Aptitude, 12 Technical MCQ, 2 Problem Solving Questions

Areas covered: Aptitude – Quantitative, Logical Reasoning, Data Interpretation.
Technical MCQ – Output of pseudo code, Find the error, MCQ.
Problem Solving – HashMap, Graph. (2 questions)

Your Comments: It is a medium difficulty round. We will be having time limit for each section in this round. Aptitude (2 mins per question), Technical MCQ (90 seconds per question), 2 Coding questions (20 minutes + 30 minutes). We will be having negative marks of 33% for each wrong answer in aptitude and technical MCQ section. So, managing the time will be difficult in this round. Coding questions are of medium difficulty level.

Have you cleared the round : YES
Details about Questions on 1st round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
DS & Algo (HashMap)	Given a set of queries, where each query is to either return a value of the key or add the key, value pair in a dictionary which is a cache with a timer for destruction of the cache.	Maintained the cache properly. Whenever it is destroyed, we should properly re-initialize and set the new time for destruction.	
DS & Algo (Graph)	Given a graph, we should return the nodes of an edge which will separate a graph into two sub-graphs if removed.	Traverse the adjacency matrix of the graph and find the cell which has only one adjacent value of 1 near it and return both the nodes.	

ROUND 2: Technical Round I (90 minutes)

Platform: HackerRank – Code Pair

Round details: I had a couple of members in the panel for the interview. I was asked to intro myself and was given 2 problem solving questions to solve it with the best approach.

Areas covered: Data Structures and Algorithms, Operating Systems, Probability.

Your Comments: The interviewer expected the best solution for the problem given. I was asked only the approach of the problem. So, try to optimize the solution as possible as you can.

Have you cleared the round : YES
Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
DS & Algo	Find the minimum element in a sorted and rotated array.	Modified Binary Search	https://www.geeksforgeeks.org/find-minimum-element-in-a-sorted-and-rotated-array/
DS & Algo	Search an element in an array and return its index where difference between adjacent elements in array is 1.	Modified Linear Search	https://www.geeksforgeeks.org/search-an-element-in-an-array-where-difference-between-adjacent-elements-is-1/
OS	Given a scenario that RAM is 4GB and Secondary memory is 8 GB. How to achieve the virtual memory for a game of 8 GB to run without lag.	Paging, Brief explanation of Data Structures like Priority Queue, HashMap used to handle the scenario with its respective complexity analysis.	
Probability	Conditional Probability	Bayes Conditional Probability	

ROUND 3 : Technical Round II (60 minutes)

Platform: HackerRank – Code Pair

Round details: I had a couple of members in the panel for the interview. I was asked to intro myself and was given 3 problem solving questions to solve.

Areas covered: Data Structures and Algorithms, Puzzle.

Your Comments: The interviewer is satisfied with the better solution. I was asked to code only one question and approach for two questions and I was asked a puzzle.

Have you cleared the round : YES
Details about Questions on 3rd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
DS & Algo (Array)	Parking Dilemma	Sliding Window	https://www.youtube.com/watch?v=FfKRqIh57lQ
DS & Algo (Binary Tree)	Minimum number of cameras required to monitor all nodes of a Binary Tree.	Level Order Traversal	https://www.geeksforgeeks.org/minimum-number-of-cameras-required-to-monitor-all-nodes-of-a-binary-tree/
DS & Algo (Graph)	Predators and Species	Modified DFS Traversal	http://prochal.com/2019/06/the-jungle-book/
Puzzle	Pay an employee using a gold rod of 7 units.	Cutting the gold rod to lengths 4, 2, 1 cm and exchange the pieces accordingly.	https://www.geeksforgeeks.org/puzzle-4-pay-an-employee-using-a-gold-rod-of-7-units/

AREAS TO PREPARE :

Data Structures and Algorithms, OOPS, DBMS, OS and strong in one programming language for problem solving.

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :

LeetCode, GeeksforGeeks, HackerRank, YouTube.

OVERALL EXPERIENCE :

First round was quite difficult. Time should be utilized very efficiently to solve many questions. Coding questions had corner test cases which was difficult to figure out. I couldn't pass all the test cases in the given time. Second round was challenging. It was about 90 minutes. The objective was to find the optimal solution for almost every problem. In the light of time, I came up with a mediocre solution and improvised it. I made sure that I was thinking aloud so that the interrogator could conceive how I narrowed down to the best solution. The ask of third round was further difficult with questions ranging from complex data structures and odd puzzles. The overall experience was really good.

GENERAL TIPS :

1. Be very good in problem solving and complexity analysis. It will be the major part of the process. Also, be clear in the core subjects such as OS, DBMS, Networking.
2. Talk out what goes on your mind about the problem given to you for solving, so that the HR gets to know how your thought process is and where you face the problem and can try to help you.
3. Should be ready for the follow-up questions.
4. Don’t hesitate to clarify the problem clearly with the HR.
5. Don’t rush for the optimal solution. Explain step-by-step optimization of the naïve solution to the optimized solution.

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PLACEMENT AND TRAINING CELL

PLACEMENT YEAR 2021-2022

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SALARY (CTC)	: 35 LPA
INTERN OFFERED?	: NO
BOND	: NO
HAVE YOU PLACED?	: NO

*(Please comment this section **in detail** to guide your juniors in bright way)*

COMMENTS ON THE SELECTION PROCESS :

ROUND 1 :

Platform: hackerrank

Round details: NUMERICAL MCQs, TECHNICAL MCQs AND 2 CODING QUESTIONS

Areas covered: DBMS, GRAPH THEORY, HASHING AND SOME NUMERICAL APTI QUESTIONS

Your Comments: both numerical and technical aptitude was of the easy-medium type and coding questions were of medium type

Have you cleared the round : YES
Details about Questions on 1st round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
graph-theory	find weak links in a graph(bridge)	Tarjan's algorithm to find bridges	
hashing	caching system	using hashmap	

ROUND 2 :

Platform: hackerrank

Round details: two to one technical interview

Areas covered: Data structures and project discussion

Your Comments: This round was easier than the first round. Questions were from my areas of interest and project

Have you cleared the round : YES
Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
graph	minimum number of days to spread a virus to all the people in a matrix	BFS	
array	minimum number of changes needed to make an array zigzag (low, high,low, high)	linear traversal with some if-else	

puzzle	a function rand() return 0 or 1 with a probability of 0.5 for each. write a function newRandom() that returns 0 with 0.25 prob and 1 0.75 prob	newRandom(){
		return 1 - rand()*rand()
		}	

general	what might be the cause of an error when the same error occurs in different lines while debugging it for more than 10 times	linker error (my guess)	

ROUND 3 :

Platform: hackerrank

Round details: problem-solving and OS concepts were tested

Areas covered: OS, algorithms

Your Comments: Questions were of medium type

Have you cleared the round: NO
Details about Questions on 3rd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
problem-solving	given a sequence of characters without space and a dictionary, return a meaningful sentence	Dynamic programming + memoization	
OS	1.what is context-switching? 2.contents stored in a process control block 3.what is a thread 4.resources shared by all the threads of a process		
problem-solving	given a file of strings, find the longest string that can be formed by just appending a single character to another string example: given [a,ab,p,pde,abc,pdeg] -> return abc as ab is formed from a and abc is formed from ab. PDEG is not the answer because there is no way to form pde because there is no pd	using hashmap with time complexity O(n*(len of longest string))	
problem-solving	given a list of 0s and 1s from the index of 0 whose removal will give the longest length of continuous 1s ex : givan [0,1,0,1,1,1,1,0,1,1,1,1,0] -> return 7	I used prefix array and suffix array approach that uses extra space but, The optimised approach was the sliding window approach	

Puzzle: given 12 objects, each of weight w except one object whose weight will be lesser than or greater than w. Given a weighing machine. find the odd one out with only 3 attempts to use the weighing machine (I was not able to answer this one)

AREAS TO PREPARE :

Data structures and Algorithms
Operating systems
database management system

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :

Geeks for geeks (problem-solving and technical concepts)
Abdul bari youtube channel (data structures)

OVERALL EXPERIENCE: The interview process was smooth and the questions were easy and answerable

GENERAL TIPS :

1. Be confident and try to answer without getting nervous
2. Don’t approach the problems in a complex way expecting the question to be very hard. Start from the naive approach
3. Be strong in Data structures and algorithms.

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PLACEMENT AND TRAINING CELL

PLACEMENT YEAR 2021-2022

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SALARY (CTC)	: 34lpa
INTERN OFFERED ?	: YES / NO
BOND	:
HAVE YOU PLACED ?	: NO

*(Please comment this section **in detail** to guide your juniors in bright way)*

COMMENTS ON SELECTION PROCESS :

ROUND 1 :

Platform: hackerrank

Round details: level = easy

Areas covered: general and technical aptitude

Your Comments: as it carries negative marks answer only if you're confident enough.

Have you cleared the round : YES
Details about Questions on 1st round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Query based	Cache functions like set, get values	using hashmap write different functions accordingly	
graph	find the cut edge of graph	dfs	

ROUND 2 :

Platform: hackerrank

Round details:level = easy

Areas covered: dsa

Your Comments:

Have you cleared the round : YES
Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
binary search	search for an element in 2d sorted array	binary search	
puzzle	number bulbs among 100 after turing on and off on multiples of N integers till 5		
	there was an another coding question, i forgot		

ROUND 3 :

Platform:hackerrank

Round details:level = medium

Areas covered: competetive programming

Your Comments:
Problems were not direct at any point. Try to observe the core idea and map it with your existing knowledge

Have you cleared the round : NO
Details about Questions on 3rd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
dynamic programming	variation of knapscak problem	using 1d dp array	
	Find the degree of array(maximum frequency of elements) and minimum length subarray that has atleast one element repeated to the frequency of degree. ip: [1,2,1,1,2,4,2] op: degree = 3, subarry [1,2,1,1]	using hashmap find degree. use the same to store starting index and ending index, concurrently update end index and minimum length	

AREAS TO PREPARE :
Competetive programming
time complexity analysis
basics of academic subjects

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :
geeks for geeks
leetcode
takeUforward,Abdul bari youtube channel

OVERALL EXPERIENCE :great

GENERAL TIPS :
Have a good undertstanding on your project but mainly choose cool projects.

Don't memorize coding techniques. Your approach towards the problem is more important than coding correct solution.

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PLACEMENT AND TRAINING CELL

PLACEMENT YEAR 2021-2022

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SALARY (CTC)	: Rs.34,25,000 - 35,25,000
INTERN OFFERED ?	: NO
BOND	: NIL
HAVE YOU PLACED ?	: NO

COMMENTS ON SELECTION PROCESS :

ROUND 1 :

Platform: Hackerrank

Round details: Aptitude round

Areas covered:

Probability, Speed – distance, Logical reasoning, Data interpretation, Data structures, SQL queries – output, Guess the output, Debugging

Your Comments:

Comprised of general, technical aptitude and 2 coding questions. Time limit were given for each section. +3 marks for every correct answer and -1 mark for every wrong answer. You can't move from one section to other in the left over time. There were nearly 15 testcases for each programming question and can be unlocked to find the error. We were allowed to choose programming language of our choice.

Have you cleared the round : YES
Details about Questions on 1st round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
2d array	Given a 2d array implement the given operations of get and set.		
Graph	To find the transaction center of a graph if any. Transaction center is the edge on removing which the graph becomes disconnected. **Sample Input:** 4 [1 2, 1 3, 2 3, 1 4] **Output: 1 4** (If we remove 1 4, we can't reach 4)	For each edge, check whether all the nodes can be reached on removing that edge from the graph. If yes, that is not the transaction center, else return that edge.	https://www.geeksforgeeks.org/check-removing-given-edge-disconnects-given-graph/

ROUND 2 :

Platform: Codepair - Hackerrank

Round details: Programming round

Areas covered: Coding & Puzzle

Your Comments:

It was a face to face interview. There were 2 HRs present and they gave programming questions one after the other and expected us to give correct approach first and then code it. They were very encouraging and also gave some ideas to me for solving the puzzle. Every time, they asked me about my understanding of the question and also cleared all my doubts.

Have you cleared the round : YES / NO
Details about Questions on 2nd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
Linked list	The nodes of the list contains binary representation and asked to find its decimal number. **Sample input:** 4, 1 -> 0 -> 1 -> 0 **Output:** 10	Normal binary to decimal conversion. (1*2^3)+(1*2^1)	https://www.geeksforgeeks.org/decimal-equivalent-of-binary-linked-list/
Buy and sell	Given an array prices where prices[i] is the price of a given stock on the ith day. You want to maximize your profit by choosing a single day to buy one stock and choosing a different day in the future to sell that stock. **Sample input:** 7 1 5 3 6 4 **Output:** 5 (6-1)	Max=0. Iterate from right and find the difference between current element and maximum element from it's right subarray. Return max value	https://leetcode.com/problems/best-time-to-buy-and-sell-stock/
Puzzle	3000 bananas and 1 camel puzzle	Till we reach 2000 bananas, 5 bananas for each km. Till we reach 1000 from 2000, 3 bananas for each km. Then 1 banana for each km	https://www.geeksforgeeks.org/puzzle-15-camel-and-banana-puzzle/

Searching	Search an element from a rotated sorted array and return its index	Using modified binary search	https://www.geeksforgeeks.org/search-an-element-in-a-sorted-and-pivoted-array/
Array	Given an array, find the degree (max occurrence of any element) and return the length of subarray with all the occurrences of that element. If more than one element occurred max times, return the smallest length. **Sample input:** 4 1 1 2 2 1 3 3 **Output:** 5 (Degree – 3 (1 occurred 3 times) subarray -1 1 2 2 1)	Find the max occurred elements and for each element find the index of first and last occurrence of that element and calculate length.	https://www.geeksforgeeks.org/smalles - subarray-with-all-occurrences-of-a-most-frequent-element/
Problem solving	Given a number, find the number of ways it can be obtained by adding 1,2,4 various times. **Sample input: 5** **Output: 4** (1+1+1+1+1, 1+1+1+2, 1+2+2, 1+4)		

<u>ROUND 3</u> :

Platform: Codepair - Hackerrank

Round details: Technical interview

Areas covered: Coding, Problem solving, OS

Your Comments:

It was a face to face interview. There were 2 HRs present and they asked question from OS. After that they asked some problems based on logical reasoning, they also asked programming questions. They gave clear explanation of all queations and ready to give some more time for our thinking.

Have you cleared the round : NO
Details about Questions on 3rd round:

QUESTION DOMAIN	QUESTION	SOLUTION / HOW DID YOU APPROACH	REFERENCES (If any)
OS	Virtual memory Context switching Process Control		

	block Threading		
Problem solving	Given digits 2, 2, 3, 3, 3, 4, 4, 4, 4 how many distinct 4 digit numbers greater than 3000 can be formed?		https://www.geeksforgeeks.org/gate-gate-cs-2010-question-65/
Problem solving	There is a pile of 12 coins, all of equal size, but only 11 are of equal weight. Can you find the unequal coin and determine if it is heavier or lighter, in 3 weighs?		http://puzzles.nigelcoldwell.co.uk/one.htm
Recursion	Given an input string and a function isvalid(word) which returns true if the given word is valid, divide the given string to words. **Sample input:** ilikecoding **Output:** [i, like, coding]		https://www.geeksforgeeks.org/word-break-problem-dp-32/
Array	Given a binary array, find the index of 0, on changing which will give maximum length subarray of 1. **Sample input:** 1 0 0 1 1 0 1 **Output:** 5 (Replacing array[5] to 1 will give length 4)	Find the maximum length subarray of 1 which has only one zero and return its index.	https://www.geeksforgeeks.org/longest-subsegment-1s-formed-changing-k-0s/

AREAS TO PREPARE :

- Quantitative aptitude
- Logical reasoning
- Problem solving
- SQL
- Data structures and Algorithms
- Operating Systems.

SITES / BOOKS YOU SUGGEST FOR PREPARATION FOR THE PROCESS :

- Leetcode
- Hackerrank
- Geeksforgeeks
- Indiabix

OVERALL EXPERIENCE :

First round was slightly difficult and solving those required more time. Time has to be managed efficiently to answer more questions. In the second and third round, they focused on our approach more than implementation. Overall experience was good.

GENERAL TIPS :

- Practice coding daily – Solving more number of questions will help in solving within the given time
- Manage time efficiently – Though you know how to solve, how you solve within given time is important
- Start with the brute force approach and then optimize your solution.
- Focus more on reducing space and time complexity.
- Ask for clarifications whenever required
- Don’t panic